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IoT-Based Food Quality Monitoring System

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ABSTRACT

The demand for fast, precise, and sensitive food safety methods is growing as consumers increasingly rely on online food delivery services. Food products shipped from South America to Europe can spend more than 21 days in transit, often resulting in deterioration, mould, or pathogen growth. Similar risks apply in the food service industry, where food may have been stored or handled improperly, leading to foodborne illness. This paper presents the Edispotter, an IoT-based food quality monitoring system designed to address these and similar issues. Using Raspberry Pi and ESP32, the Edispotter collects essential food quality data through various sensors. The data are processed and stored in a Redis database within an Amazon Web Services (AWS) cloud environment, providing real-time food-based status updates via an Android application.

1 | Introduction

Proper food safety standards are essential to avoid spoilage and preventable foodborne illness. Modern food transportation networks, such as ocean freight, can see foodstuffs in transit for up to 30 days. Although some steps have been taken to detect mould growth in transit [1], such a system is not always efficient [2], requiring other methods. Bananas generate ethylene as they age, which can cause ethylene gas-induced ripening in other fruits [3]. One solution is to monitor containers used in banana storage via sensors designed to detect ethylene and infer the freshness of the food contained within [4].

Other studies have shown that environmental factors have a pronounced impact on the freshness of meat products during production, a marker of spoilage is the emission of ammonia (NH_3) due to microbial action [5]. Using a sensitive NH_3 sensor, the level of spoilage can be inferred before visible changes. One such device is an adaptable NH_3 sensor using polymer poly (3,4-ethylene dioxythiophene): poly (styrenesulfonate) PEDOT: PSS/

silver nanowire (AgNW) composite film as the dynamic layer. With AgNWs consolidated into the PEDOT: PSS film, sufficient sensitivity can be achieved [6] for use in a sensing system to determine the extent of meat spoilage.

Internet of Things (IoT) devices have been used to monitor important factors in the spoilage of raw milk, such as temperature and quantity of micro-organisms present [7]. Such systems allow for machine learning models to actively monitor dairy facilities, improving the efficiency of the operation [8].

Although the true origin is still unknown, the suspected source of the ongoing Severe Acute Respiratory Syndrome Coronavirus-2 (SARS-CoV-2) pandemic, commonly known as COVID-19, has been linked to a wet-market in Wuhan, China. It is believed to have spread from contaminated meat [9], emphasising the benefit of adequate food monitoring.

This paper presents a unified IoT system for monitoring the indicators of food spoilage. Specialised sensors are used to

This article relates to wireless sensor networks and IoT.

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measure the conditions inside a test chamber, such as temperature and humidity, as well as markers of food condition, such as Air Quality Index (AQI), food temperature, and Potential of Hydrogen (pH). The system is built on a Raspberry Pi Model 3B+ and ESP32 wifi capable microprocessor, the data are transferred over the network via Message Queuing Telemetry Transport (MQTT) to Amazon Web Services (AWS) for storage, and finally to one or more Android devices for display.

2 | Related Work

Studies have investigated the use of sensors to monitor food quality inside IoT smart-refrigerators [10, 11] focusing on both domestic and commercial benefits. Other studies in the dairy industry have demonstrated the benefit of monitoring unpasteurised milk, which may contain pathogens or micro-organisms. In one study [12], a machine learning model determines the state of raw milk before it moves to other stages of production. The system uses various gas sensors [13] monitored by an Arduino microcontroller, sending via local network. The data is processed then presented to a user via Android mobile device. While such a system is suitable for liquids, it may not be suitable for other food items as it only focuses on gas sensing, and there may be other indicators of food spoilage. Furthermore, this study utilises an Arduino which offers less flexibility than the more feature rich Raspberry Pi.

Milk is a profoundly short-lived item [14], Studies [15, 16] determine milk quality based on its bacterial content, where the temperature and quantity of micro-organisms are measured. This informs to which vessel a given milk sample will be sent for storage based on what standard the milk qualifies. The decision is made by a machine learning model using sensor data as inputs. Although highly suitable for industrial settings, it is inappropriate for the consumer market due to scale and cost. Although this system focuses on bacterial content and the indicators thereof, pH, ambient temperature and humidity, and air quality of the storage vessel have an impact on the quality of milk [17]. This implies that more sensors can be employed to create a more holistic representation of condition of the milk.

The primary cause of meat spoilage is bacterial action during production, transport, and storage. There is an economic benefit in reducing loss to spoilage by monitoring the state of meat products, such as through NH_3 production [4]. One study monitored NH_3 sensors using PEDOT: PSS/silver nanowire (AgNW) composite film as the dynamic layer and a data acquisition system to monitor the freshness of meat indicated by NH_3 measurements [6, 18]. PEDOT is not readily available outside of laboratory procurement and the process described in the paper is time consuming. Another used a Convolutional Neural Network (CNN)-based electronic nose system, mimicking mammalian olfactory sense to detect odour [19]. Drawbacks to such a system is that odour may be relatively ambiguous when determining spoilage state versus NH_3 production.

Traditional storage methods involve centralised servers, however, block-chain technology is an innovation that allows for

decentralised and secure data storage [20]. One study presented a quality monitoring system for fruit juice production [21]. During production smart contracts are executed to record data on a block-chain. Based on measurement results, the smart contracts will determine if the juice production will continue in that stage of the process. However functional at a local level, blockchain has not yet become the revolutionary force it is claimed to be and has not transformed supply chain activities. Benefits to a block-chain based system are that modern IoT devices such as Radio Frequency Identification (RFID) tags or other sensors can be integrated into a unified block-chain-based system [22]. More recently, one study utilised block-chain technology to implement a fruit and vegetable tracking system via 'double chain and double storage' blockchain model. It also uses smart contract technology for filtering and restriction of data upload [23] meaning similar issues will persist.

Safe drinking water is essential for good health and well-being [24]. Monitoring water quality involves collecting various data streams at standard intervals [25, 26]. The physiochemical parameters of water quality are represented as temperature, pH, conductivity, and oxidation-decrease potential [27, 28]. The study's sensors use the Zigbee standard [29] where transmitter modules send data to 'estimating' and 'notification' nodes. Notification nodes produce an audible alert when water quality parameters are outside acceptable levels. The primary drawback to this form of sensing is that the Zigbee standard has a relatively low data rate and range when compared to other connected sensors using conventional WiFi or wired connections. Additionally, the system required a high degree of physical maintenance [30].

Commercial baking of sponge cakes requires various quality indicators. One study employed a single sensor hyperspectral imaging (HSI) approach to simultaneously quantify the cake's moisture content and hardness. HSI is a spectroscopic imaging method that captures spatial and spectral images [31]. HSI imaging systems capture multiple images in different wavelengths of the electromagnetic spectrum [32], however, in the study they found that models perform better for moisture measurements in white sponge cakes compared to those of other colours such as chocolate. Aside from this limitation, the cost and complexity of the system is quite high, requiring large data storage capacity as a single image is a large, multidimensional dataset, hundreds of megabytes in size. Similarly, a study using an IoT suite of sensors was employed in a commercial bakery for monitoring environmental factors, such as ambient dust, temperature and humidity. This study made use of a DHT22 sensor for temperature and humidity, and a GP2Y1010AU0F optical sensor to monitor airborne particulate. Both sensors communicate through a wireless network via ESP32 WiFi module to provide alerts to an administrator. Although the system functions well for this exact task, it does not directly monitor the food items [33]. Knowing that dust can accumulate in an environment, a persistent subthreshold amount of dust may accumulate and become problematic. Knowing this, the status of the food items in question would be an additional benefit to such a monitoring system.

Monitoring defrosting food is crucial to ensure food safety, as unintentional thawing can lead to bacterial growth, and

compromised food safety. This study used microwave frequencies emitted via Ultra-Wide Band (UWB) radio transceivers to measure the refractive index of frozen food as it thaws [34]. However, using UWB in such a configuration requires a controlled environment to achieve an accurate reading, and use of the system is limited to use inside the cooler. In addition, a cell phone is stored inside the cooling unit, where its camera is used to assist in monitoring the sample. The use of a cell phone over a dedicated camera unit may represent a mismanagement of resources, as modern phones are highly capable portable computers that may better suited to more complex tasks. A portable sensing device would allow for an additional number of functional environments.

Transportation of food ordered via delivery apps has become commonplace since the adaptations inspired by the COVID-19 pandemic. Food safety between ordering and delivery is ultimately the responsibility of the delivery drivers. This study proposed a system comprising a series of sensors that communicate via WIFI to monitor the quality of food in transit. In particular, they monitor the temperature of food using a temperature sensor to determine its freshness, a Global Positioning Satellite (GPS) tracker is used in conjunction with a servo motor serving as a container lock to ensure that the food is not tampered with in transit. The food is only unlocked and available upon arrival [35]. The primary drawback of this system is that the only quantity being monitored is temperature; this means that if food that was contaminated, or spoilt before leaving the source, it would likely still be consumed on arrival. This system, while innovative, leaves room for one that performs more holistic monitoring.

To overcome the shortcomings in these studies methods, while capitalising on the advantages presented in each paper, a multi-sensor system has been devised. Utilising a pH [36], temperature [37], humidity [38], light [39], and air quality [40] sensor, monitored by a Raspberry Pi [41, 42]. The data will be sent via network to AWS using MQTT protocol [43, 44]. The most up to date data will be displayed on an Android device [45].

3 | The Edispotter System

According to the Center for Disease Control and Prevention [46], an estimated 48 million people will contract a foodborne illness each year. Data show that the bulk of the infections are caused by individuals consuming poorly inspected or expired foods. In industrial settings, food freshness is assessed using quality assurance systems, while at home, individuals must evaluate freshness themselves [47]. The techniques used in the industrial environment may applied to the domestic market following a sufficient reduction in scale, through the use of small-scale consumer electronics.

This need facilitates the Edispotter, an IoT system based on a Raspberry Pi 3B+, various sensors, and an ESP32 module. Figure 1 shows the architecture of the Edispotter. Measurements take place in a closed container with a semi-permeable gasket which allows for gas exchange, enabling the measurement of environmental factors such as AQI and humidity but

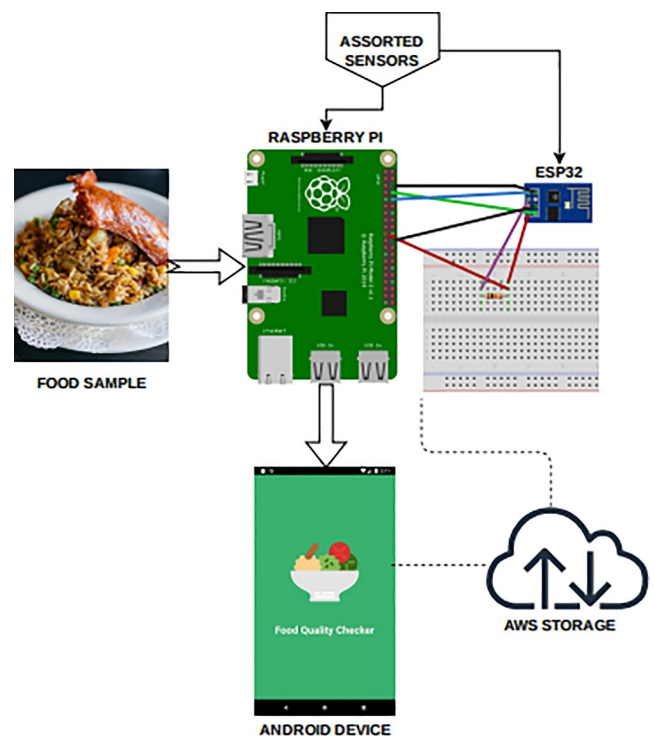


FIGURE 1 | Architecture.

preventing the accumulation of humidity, as it may interfere with gas sensors. First, a sample is placed in the test chamber, then data is collected via various sensors measuring pH, AQI, temperature of the chamber, temperature of the food and ambient humidity. This data is processed via Raspberry Pi, and transmitted via ESP32 to Redis-AWS where data analysis takes place, finally an android application displays this analysis to an end user.

The prototype developed during this study addressed the problem statement and addresses two contrasting scenarios.

- Scenario 1: The entire model follows the execution presented in Figure 1. The user receives the food, the whole meal sensed using a food monitoring device, values are passed to a mobile Android device enabling the user to view the status of food.
- Scenario 2: The model is enhanced, providing a dashboard for all the dishes at a given restaurant, enabling the customer can select a dish based on its status in the dashboard.

The prototype was powered by a Raspberry Pi 3B+ [48], which monitored sensors and handled Input/Output (I/O) operations. Sensor data was sent from the ESP32 over the network to the AWS Central Data Repository (CDR) for processing, and could then be accessed via the user's Android device. All software was written in Python 3.5 or higher.

Although the device can be configured to use many different sensors given the appropriate hardware support and signal conditioning, the sensors used in this study were selected for their accuracy, cost-effectiveness, ease of use, and existing

community support. Figure 2 shows the sensors used during the study.

- **Temperature:** To capture ambient temperature and humidity, a DHT11 integrated temperature and humidity sensor was used. For most food products, a specific temperature and humidity are required to maintain freshness. For sensing the temperature of the food sample directly, a LMT87 Analogue temperature sensor in TO-92 package was used.
- **Light:** A TSL2561 light sensor was used to measure the brightness of the food sample, reporting Lux, Broadband, and Infrared light. As the food sample changes in colour and albedo, the sensor detects these differences. Although colour is not directly measured, these values can be inferred to show significant differences between fresh and spoilt food.
- **Air Quality:** To measure AQI, a MQ135 sensor was used. It is sensitive to NH_3 , alcohol, and CO_2 gas present in the air. The sensor is known for its long lifespan, affordability, and requires a simple drive circuit. One drawback is its poor performance in high humidity.
- **Ammonia gas:** The MQ137 gas sensor was selected to measure NH_3 directly. The sensor has a dual signal output, an analogue value which changes in proportion to the amount of gas detected, and a TTL level logic level used to trigger interrupts.
- **pH Level:** To measure pH an Atlas Scientific 'Consumer Grade pH probe' and 'Surveyor Analogue pH Sensor' were used. The probe is fully waterproof, able to operate up to 50°C , and the sensor board output is monitored by the Raspberry Pi.

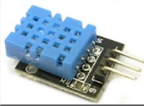
Sensor Image	Sensor Name
	DHT11 Temperature and Humidity Sensor
	LMT87 Temperature Probe
	TSL2561 Visible and Infrared Light Sensor
	MQ135 AQI Sensor
	MQ137 Ammonia Sensor
	Atlas Scientific "Consumer Grade" pH probe and Analog Sensor

FIGURE 2 | Sensor names.

Analogue devices are connected to an MCP3008 Analogue to Digital Converter (ADC) addressable via Serial Peripheral Interface (SPI), or ESP32, and monitored from the Raspberry Pi.

Once the data has been collected, the parameters are compared to standard threshold values in the Android App and presented to the user, informing them if the food under test is safe to eat.

3.1 | Architecture

Figure 1 shows the architecture of the Edispotter.

- **Food Sample:** The food sample serving as the input can be raw or cooked.
- **Assorted Sensors:** Major environmental factors drive the spoilage of food samples such as temperature, humidity, Air Quality, pH, and so on. Individual sensors have been used to detect such factors.
- **Raspberry Pi 3B+:** Performs tasks like data collection, data analysis, data transfer, etc.
- **ESP32:** A microcontroller used to convert Analogue signals to digital, and send data via MQTT/HTTP to AWS.
- **AWS Storage:** Data obtained from different sensors is stored on the cloud and analysed.
- **Android Device:** Used for data visualisation purposes, and as the front-end of the Edispotter via Android App.

Below are some of the internal processes of our Edispotter:

- **Data collection:** Temperature, light, ammonia, pH, and Air quality parameters are collected via sensors and monitored via Raspberry Pi and ESP32.
- **Data Transfer:** Messages are transmitted via Message Queuing Telemetry Transfer (MQTT) and Hypertext Transfer Protocol (HTTP) once the inputs are processed.
- **Data Storage:** AWS cloud storage powers the Redis database, and is used to store data using the key/value method.
- **Data analysis:** Comparing the measured sensor values to the predefined, unsafe threshold values.
- **Front-end visualisation:** The android app displays results and figures to the end user.

Python was used to create the web server on the Raspberry Pi. Additionally, RESTful services were used for the execution of micro-services. As shown in Figure 1, multiple sensors are connected to the Raspberry Pi to monitor food quality. Temperature, humidity, colour, pH, air quality, and ammonia are essential factors for any food or beverages. The values from the sensors are transferred through MQTT and HTTP protocols to AWS storage.

Finally, the stored data in AWS are compared to threshold values to ensure accurate results. For visualisation purposes, an Android application has been implemented, simplifying access for end-users.

3.2 | Flow Diagram

Figure 3 shows the logic flow as follows:

- **Food Sample:** Any food product may be used. In our case, a hard-boiled egg, a 2cm² section of grilling steak, and a half orange are used in the study.
- **Assorted Sensors:** Typically, five inputs are used, temperature, humidity, light, AQI, and pH.
- **Raspberry-Pi:** Sensor input and processing achieved using the Raspberry Pi. A Secure Digital (SD) Card is used for nonvolatile storage, with video output connected to an Android device.
- **ESP32:** The ESP32 is responsible for converting the pH sensor's Analogue output to digital. It additionally provides WiFi connectivity for transferring data.
- **MQTT/HTTP:** Data transfer protocol MQTT/HTTP is used to transfer data from hardware to the database.
- **Redis-AWS:** AWS offers an in-memory data store and cache service as a Platform as a Service (PaaS). The Redis database

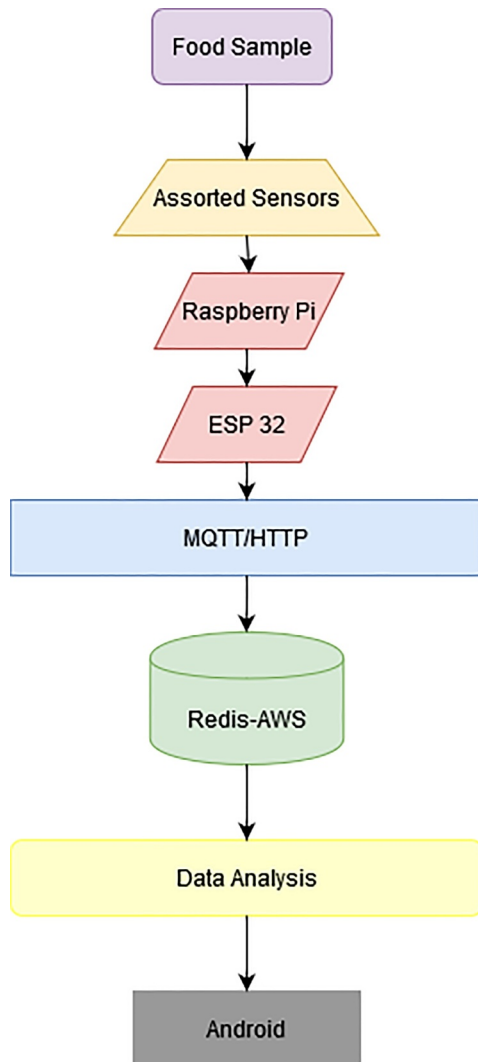


FIGURE 3 | System flowchart.

is used to store data in a key/value format for the in-memory cache.

- **Data Analysis:** Data are retrieved from the database and decisions have been made as described by Algorithm 1. Data are then visualised for the end user.
- **Android:** Finally, the processed data from all sections are pushed to the Android app which provides cumulative analysis of the various factors for the given food sample.

As shown in Algorithm 1, two parameters are considered: the actual value and the predefined value for a given food product. The primary purpose of using these two parameters is to assess the status of the food. If the measured value exceeds the threshold value, negative feedback is provided which indicates the food is not edible, otherwise, positive feedback is given to indicate the food is safe to eat.

Once the connection is established, the sensors are measured and the readings are stored in the database. The sensor data are then displayed on the Android device.

ALGORITHM 1 | Algorithm of edispotter.

```

Request from Device;
while True do
    Check Sensors;
    if Sensors ≠ True then
        Sensor Error;
    else
        Start the Sensor;
while True do
    Readings measured;
    Activate sensors;
    if Sensor False then
        Take readings;
        if Readings = True then
            Insert into Redis database in
            AWS;
        else
            Reading error;
    else
while True do
    Parameters Analysis;
    if Actual Value < Predefined Value ||
       Actual Value > Predefined Value then
        NOT Edible food;
    else
        Actual Value == Predefined Value
        Edible food;
  
```

Existing System	Flaws in Existing System	Proposed Edispotter System
1) MyIPM	- Fails to monitor fruit, only provides spoilage related information - Limited data storage	- Scans food in real-time, with the ability for display - Able to sense fruit, dairy, meat, etc. - Ample storage via AWS.
2) Open Food Facts	- Requires scanning a barcode. - Gives insufficient information to the user. - it does not indicate the suitability of the food for consumption.	- Requires no barcode data. - Provides food safety feedback.
3) Food Standards and Quality Control	- No user interaction. - Does not provide real-time information. - Data can not be added manually.	- Ample user interaction - provides real-time data and user feedback via Android App. - Optional manual data entry.
4) Yuka	- Slow response time. - Difficult and time consuming as all data must be entered manually.	- Fast response time. - Automatic data entry via sensor input.

FIGURE 4 | Differences between existing systems and the Edispotter.

Figure 4 highlights the flaws in existing systems and presents the features of the Edispotter to address these weaknesses. MyIPM [49] is an Android application designed to provide information to prevent fruit spoilage. Open Food Facts [50] is a system that scans the Universal Product Codes (UPC) of food items, but it does not support items without barcodes. Food Standards and Quality Control is a noninteractive system that provides information about food standards and quality. Yuka [51] is a dual-purpose scanner application for food and cosmetics. This system scans food items and offers information on how to prevent spoilage.

3.3 | Limitations

1. Currently, the Edispotter does not support iPhone users because it is developed as an android application.
2. A WiFi connection is needed to transfer the data from raspberry pi and ESP32 to the Cloud server for the analysis.

4 | Experimental Results

4.1 | Hardware Components

The Raspberry Pi 3B+ features a quad-core 64-bit processor operating at 1.4 GHz, with 1 GB of LPDDR2 SRAM, and Bluetooth 4.2/BLE. The compact size of the Raspberry Pi makes it portable and convenient to implement, with the functionality of a low power Linux desktop computer but at a fraction of the cost. However, the Arm core processor cannot run X86 operating systems or software, and while it can function as a personal computer, it cannot fully replace one.

The ESP32 was employed for the pH sensor as an ADC and data transmission device. It is important to note that the ESP32 is a microcontroller able to convert analogue voltage levels to digital signals, and has integrated WiFi and dual-mode Bluetooth. A notable issue with the ESP32 chip is that it cannot reliably reset to a predictable initial state, and the firmware occasionally crashes as the device is sensitive to even minor voltage fluctuations.

The DHT11 sensor was used for measuring temperature and humidity, it is a popular and low-cost digital sensor. While it is relatively easy to use, it requires precise timing to obtain accurate data. One limitation of the DHT11 sensor is that it can only provide new data every 1–2 s.

The TSL2561 light sensor was used for detecting the luminosity of food. This sensor features a patented dual-diode architecture and an Inter-Integrated Circuit (I2C) digital interface. It is sensitive to both Infrared and visible light, able to distinguish from darkness to direct sunlight.

The concentration of ammonia gas is typically high in spoiled food. The MQ137 ammonia gas sensor was used to assess food freshness, offering excellent sensitivity to NH_3 gas, along with a long lifespan, low cost, and simple driving circuitry. It is commonly used in household NH_3 alarms, industrial detectors, and portable sensors. While the sensor is not highly accurate for precise ppm measurements, it is effective for detecting NH_3 presence rather than exact concentrations. A drawback is its long warm-up time.

Air quality is a crucial factor in maintaining proper food hygiene. In this experiment, the MQ135 AQI sensor was used, which is capable of measuring NH_3 , NO_x , alcohol, benzene, smoke, and CO_2 . The sensor offers a fast response time, high sensitivity, and a long lifespan at a low cost, similar to the MQ137. However, it has limitations, such as susceptibility to false alarms and reduced performance in areas with high CO and CO_2 levels. Both gas sensors perform poorly in high humidity environments.

Every food item has an ideal pH level. The Atlas Scientific Consumer Grade Analogue pH was used to measure the pH level of food, with its Analogue output converted to digital I2C words via the ESP32. The sensor is sensitive to pH ranging from 0 to 14, and a reaction time of less than 5 s. It provides accurate results within a temperature range of -10°C – 50°C . One major drawback is that the sensor requires calibration in an acid, a base, and neutral pH solution.

4.2 | Experimental Setup

Figure 5 shows the sensor array which was mounted inside the experiment chamber. The array holds all discussed sensors and allows the pH probe (not pictured) to be held firmly in the centre. In this configuration, all sensors (MQ137 [52], MQ135, DHT11 [53], TSL2561 [54], LMT87) are securely held above the sample, with the pH and LMT87 temperature probe in contact with the food item. A 12 V LED strip was used to provide static lighting during the course of the experiment. Figure 6 shows the

experimental chamber. The chamber is blacked out with a gas-permeable fabric gasket to prevent ambient light intrusion but allow for gas exchange. The gas sensors behave poorly in a high humidity environment, so controlled gas exchange was necessary. The chamber contains the Raspberry Pi 3B+, ESP32 [55], Surveyor Analogue pH Sensor, and cabling. A repurposed 300W ATX supply was used to provide reliable current to the apparatus.

The experiment was performed with three food items, a hard-boiled egg, a 2cm² section of grilling steak, and a half orange. Each item had been cooled at 4°C for a period of 10 h prior to use in the experiment. The experiment was conducted for each item individually, followed by a separate experiment with each type of item together, also subjected to the same cooling process prior to evaluation. Each experiment was monitored for 48 h, measurements were taken once per minute, and averaged by hour. In the final experiment with three items, only gas, temperature and humidity are recorded due to limitations in the apparatus.

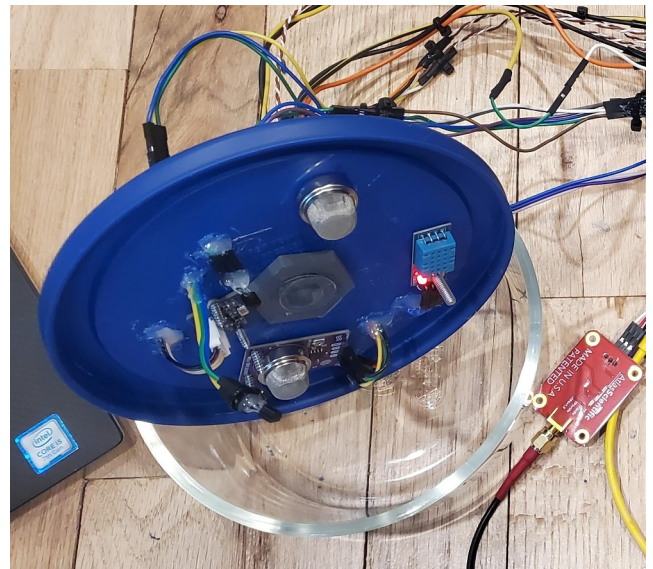


FIGURE 5 | Sensor Array and connecting cables.



FIGURE 6 | Experiment chamber containing Raspberry Pi and other devices.

Figure 7 shows the pinout of the Raspberry Pi, featuring a 40-pin General Purpose Input/Output (GPIO) header. This header includes two types of voltage pins: two 5V pins and two 3.3V pins. Additionally, there are eight ground pins (0V) to provide a common reference. A GPIO pin, when set as an output, can be configured to logic high (3.3V) or logic low (0V). These GPIO pins are further classified into four categories: PWM (Pulse-Width Modulation), Serial Pins, SPI and I2C.

Figure 8 shows the integration of the DHT11 temperature and humidity sensor with the Raspberry Pi. The three pins of the DHT11 sensor are connected to pin 2, pin 3, and pin 6 of the Raspberry Pi for input, output, and grounding, respectively.

Figure 9 shows the integration of the TSL2561 light sensor. Pins 1, 5, and 9 are used for input, output, and ground.

Figure 10 shows the connection of the MQ135 air quality sensor to the Raspberry Pi. For this sensor, pins 4, 6, and 15 are used for input, output, and ground, respectively, connected to the board.

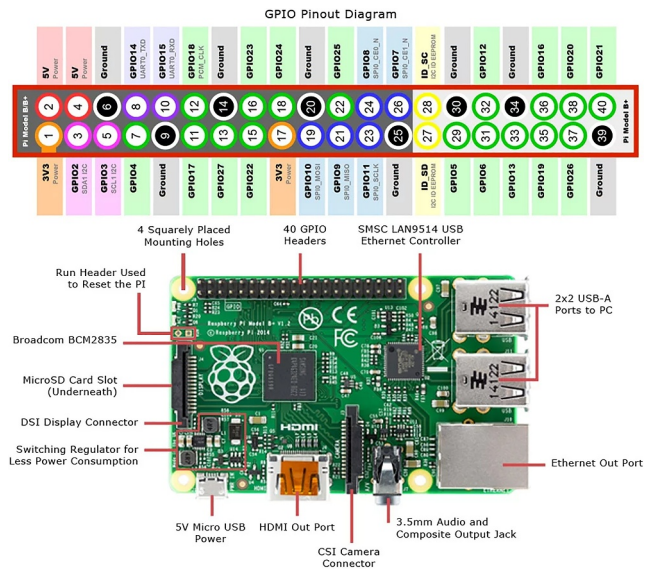


FIGURE 7 | Raspberry Pi pinout diagram.

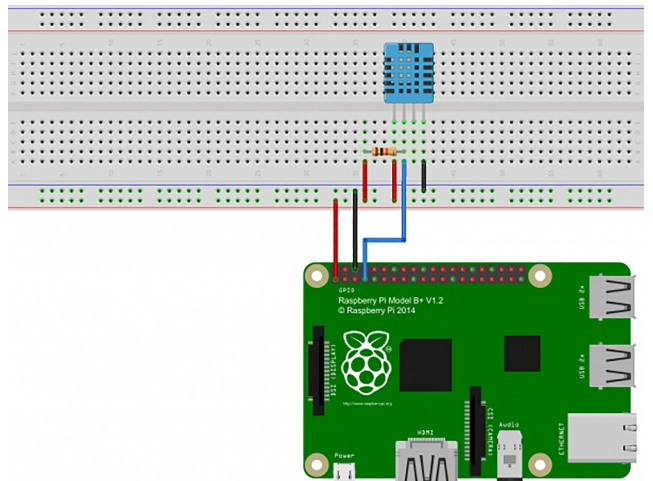


FIGURE 8 | DHT11 Temperature and humidity sensor connection.

Figure 11 shows the connection of the MQ137 ammonia gas sensor to the Raspberry Pi. In this configuration, pins 36, 2, and 39 are used for input, output, and grounding.

Figure 12 shows the connection of the ADC converted I2C pH signal to the Raspberry Pi. In this configuration, Pin 3 is configured as SDA, pin 5 as the SCL, and pin 25 as the ground.

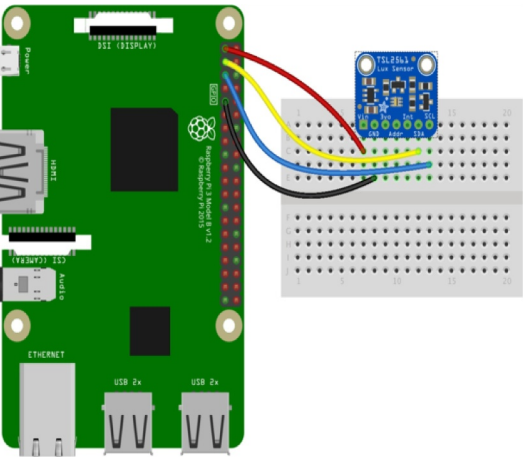


FIGURE 9 | TSL2561 light sensor connection.

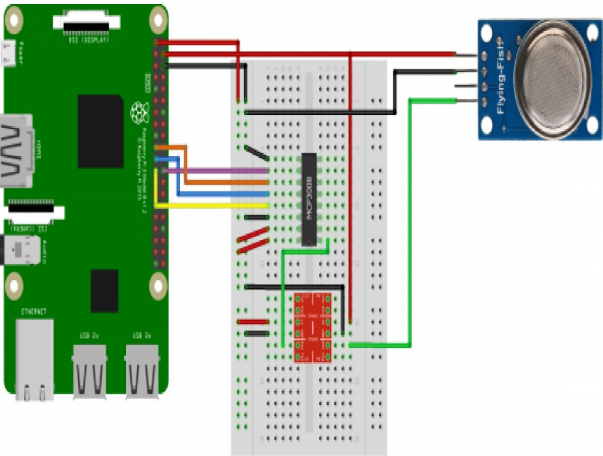


FIGURE 10 | MQ135 air quality sensor connection.

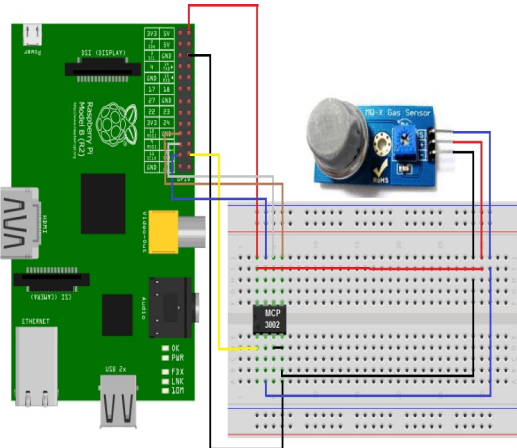


FIGURE 11 | MQ137 ammonia sensor connection.

4.3 | Performance Analysis

During this study, all experiments were run for 2 days, this was deemed adequate as most food left in ambient conditions would be considered unsafe to consume after only a few hours.

Figure 13 shows the changes in the egg temperature over time. Unsurprisingly, the egg temperature matches closely with ambient temperature after the first hour as the egg has begun to warm to room temperature. As the experiment progresses, this trend continues with the egg temperature matching more closely with ambient temperature. The temperature starts above 4°C as the preparation time and physical handling increased its temperature before data collection could begin.

We can see the meat temperature in Figure 14. It should be noted that the temperature of the raw meat as the experiment begins is already close to room temperature. Despite efforts to reduce handling time, the meat section had low thermal mass and warmed quickly. As the experiment continues, we can see

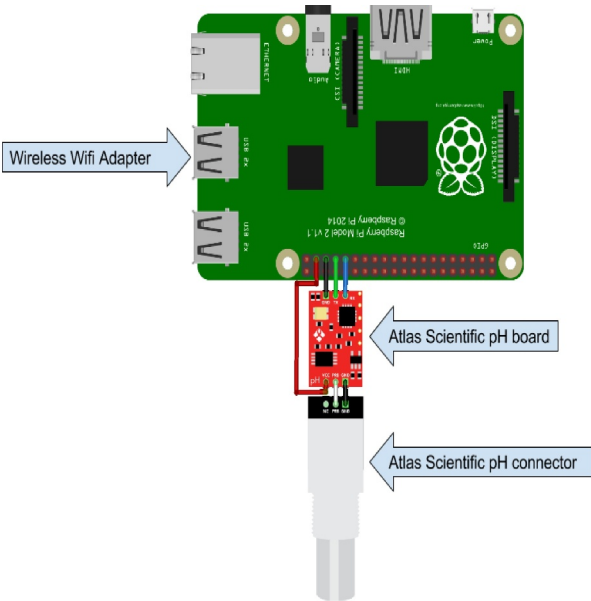


FIGURE 12 | I2C pH sensor connection.

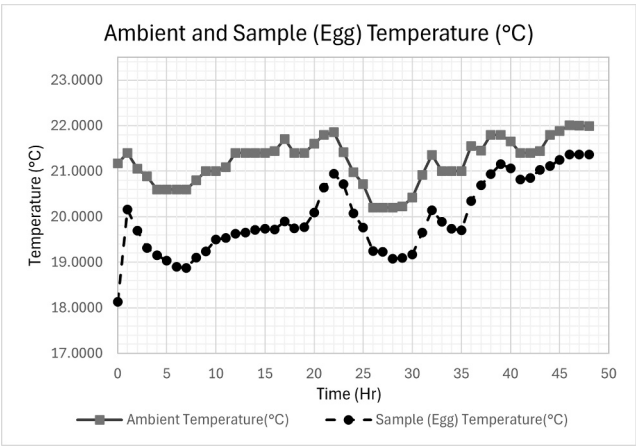


FIGURE 13 | Boiled egg and room temperature readings.

that the meat temperature slowly approaches that of the chamber.

In Figure 15, we can see that the half orange follows a more expected temperature curve. This is likely because the orange contains significantly more water than the previous food items, making it more resistant to temperature changes. In each trial, we can see that the food has effectively reached room temperature after 5 hours.

Although the humidity fluctuates during the experiment, we can see in Figure 16 that humidity starts near 61% then drops rapidly. This corresponds to the egg temperature; what was likely measured in the first 5 hours was the humidity leaving the egg and escaping through the gas-permeable gasket. The egg was seen to have dried noticeably at the end of the experiment, the fluctuations seen likely during the experiment are the result of ambient sources of humidity.

Similarly with the egg, in Figure 17 we see that humidity fluctuates over time but shows a more pronounced decrease. The decline is much slower than that of the egg, which may be due to the fat content of the steak section. The experiment ends with the humidity measured near 32%. On inspection, the meat had dried nearly completely.

What is likely seen with the orange experiment, seen in Figure 18, are humidity readings which are influenced by the environment. The orange contained the most water initially and reached room temperature more completely than the other items. Although there is pronounced environmental effects, if we focus on the minima of the graph, we can see a declining trend from 40% to 36%.

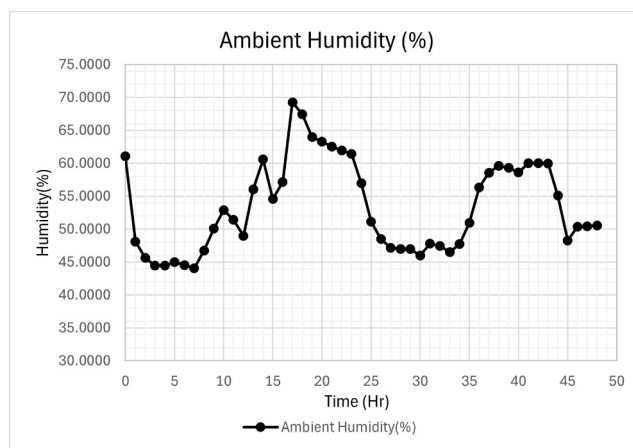


FIGURE 16 | Measured humidity during egg experiment.

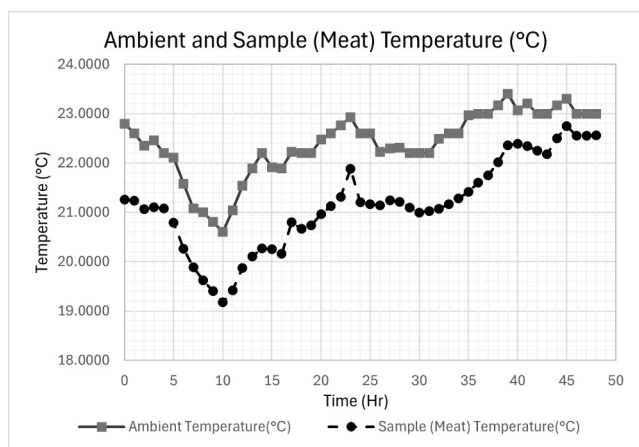


FIGURE 14 | Raw meat and room temperature readings.

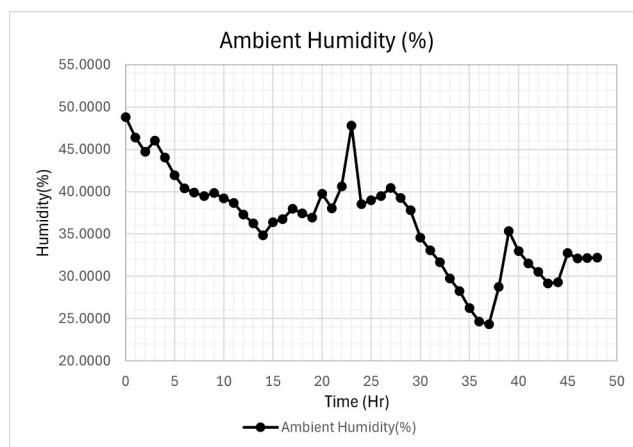


FIGURE 17 | Measured humidity during raw meat experiment.

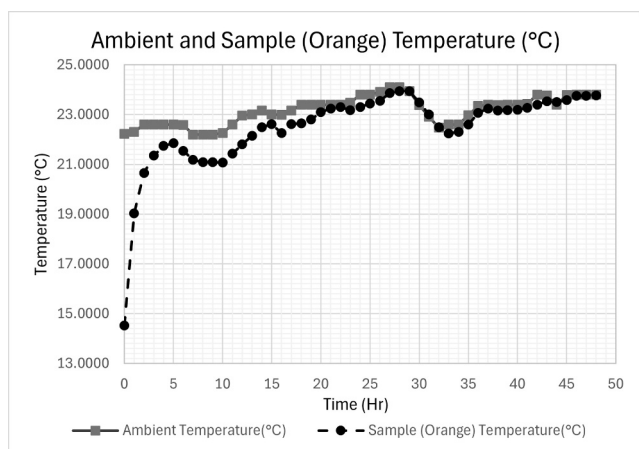


FIGURE 15 | Half orange and room temperature readings.

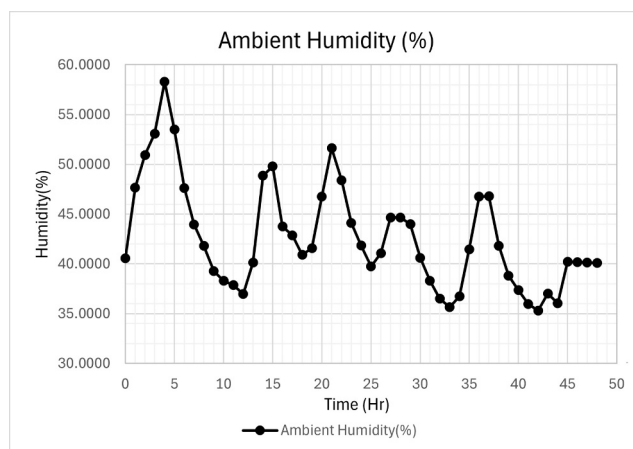


FIGURE 18 | Measured humidity during half orange experiment.

Figure 19 shows the AQI readings captured during the egg experiment. Prior to hour 18 AQI is relatively constant with a slightly decreasing trend, between hours 18 and 24 we see a wild alteration in AQI ppm. It is highly unlikely the egg had suddenly released a large quantity of gas, and more likely that the sensor had been confounded by a peak in humidity from environmental sources. A drop in humidity coincides with a return to more reasonable AQI values. Towards the end of the experiment AQI is seen to begin increasing as the egg began showing more pronounced effects of spoilage.

The data presented in Figure 20 shows the AQI ppm of the meat experiment slowly decreasing. This follows that the meat section had been persistently drying during the course of the experiment. Since drying is a method of preserving food, it stands to reason that bacterial action was slowed due to lack of moisture.

Figure 21 shows a steadily increasing AQI ppm trend, starting with one initial peak. We have seen in Figure 19 that high humidity may adversely affect the measurement obtained by the gas sensors, it is possible that this initial spike was related to the spike seen in humidity. Given that the orange maintained the highest moisture content throughout the experiment, it is likely that bacterial action is responsible for the increase in CO₂ and other gasses detected by the AQI sensor.

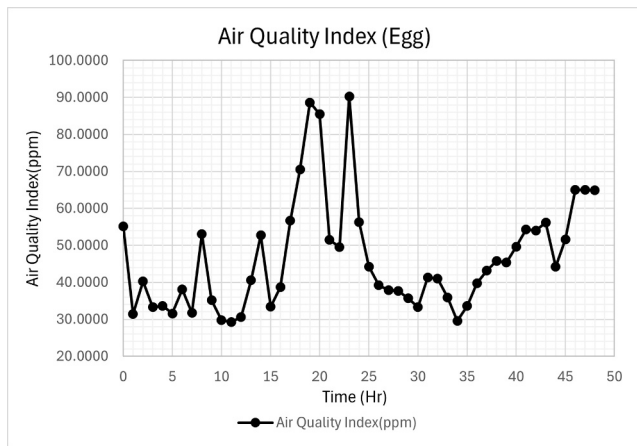


FIGURE 19 | AQI during egg experiment.

Figure 22 shows the pH change during the egg trial, starting at 8.2 and ending at 9.4. The pH probe was inserted into the egg, likely making contact with a mixture of the white and yolk, each having different pH values. Despite this, the trend reflects the typical pH increase in eggs as they age. Although it is possible that the change is due to the egg drying, and making poor electrochemical contact with the probe, the increase in pH is most likely the result of spoilage.

Figure 23 shows the meat pH rising from 6.25 to 7.25 with a small plateau at the end of the period. The plateau may be due to the meat losing moisture and having reduced contact with the probe, however we are likely seeing the changes in pH associated with meat spoilage.

In Figure 24, it initially appears that there is a great deal of change in the pH of the orange, however, the value is only changing by 0.0375 between its minimum and maximum values. The observation is likely influenced by drifts in the pH sensor, or due to electronic noise. Likely the lack of change in pH is due to the lack of change in the orange itself. What can be ascertained from these data is given by the two local minima, first at 5 hours, and again at 44 h which may indicate a slowly decreasing trend in pH as the orange ages.

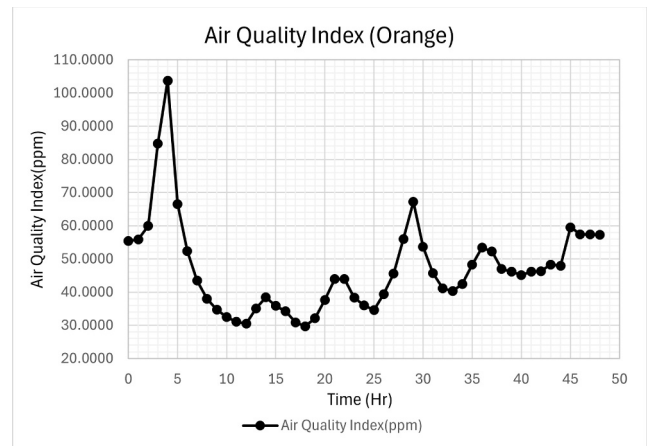


FIGURE 21 | AQI during half orange experiment.

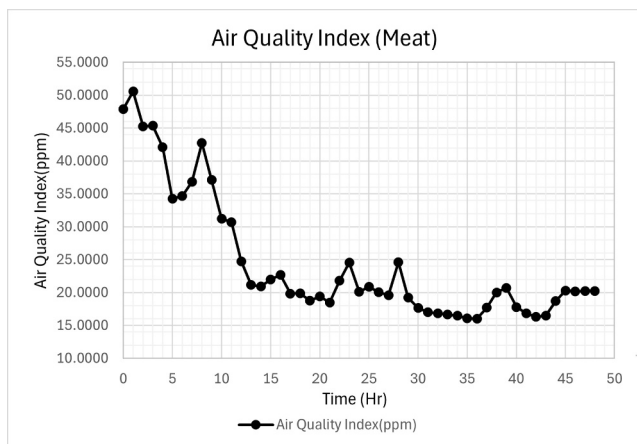


FIGURE 20 | AQI during raw meat experiment.

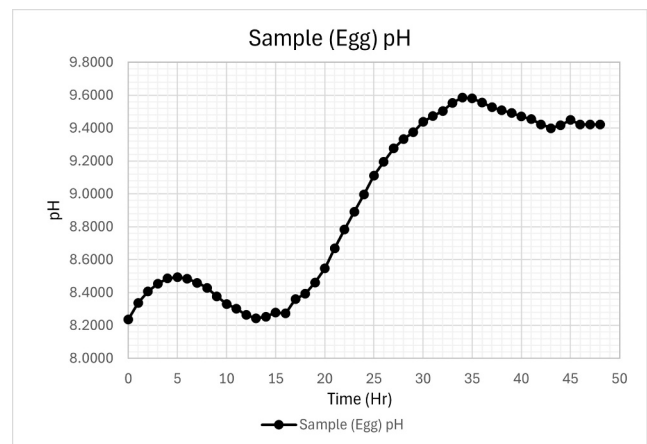


FIGURE 22 | Changes in egg pH during spoilage.

The limitations of the apparatus prevent us from measuring individual food temperatures and their pH. What we see in Figure 25 is overall increased temperature inside the chamber, as well as overall higher humidity as shown in Figure 26. This can be expected as there are three items losing moisture to the chamber. Water vapour serves as a greenhouse gas and may

have assisted in maintaining the temperature of the chamber during the experiment.

Figure 27 likely provides the best indication of changes in food during the three-food experiment. Because AQI represents microbial activity, we observe a notable upward trend after hour 20. From these experiments, it can be concluded that gas measurements, specifically AQI, may be the most indicative of food quality in a multi-food spoilage scenario, while pH, food temperature, and ambient conditions may be more useful for single-food measurements. The single-food data show that different foods produce varying levels of gas, or pH changes as they spoil, influenced by ambient conditions. If moisture is low, the food is likely to dry out, resulting in minimal changes in measured values. However, if humidity and temperature are high, AQI and pH are likely to show more pronounced changes as the food becomes colonised by bacteria and fungi.

If demands on the system were quite high, such as an entire refrigerator, the Edispotter would likely be unsuitable, since it requires specific inputs from a single food item, made evident in the final experiment using three food items. Future versions of the system could be made modular, reducing the number of inputs as the complexity of the sampling environment increases.

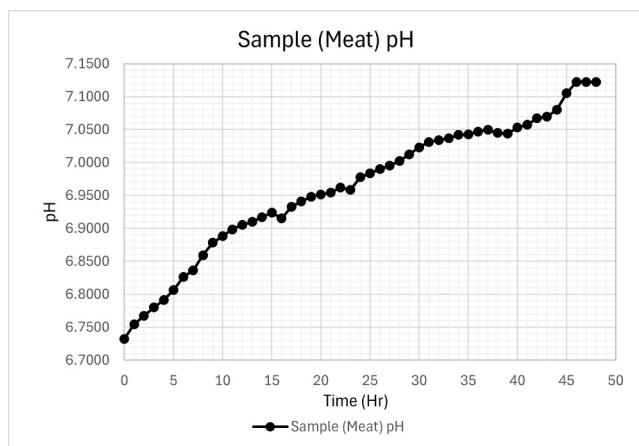


FIGURE 23 | Changes in raw meat pH during spoilage.

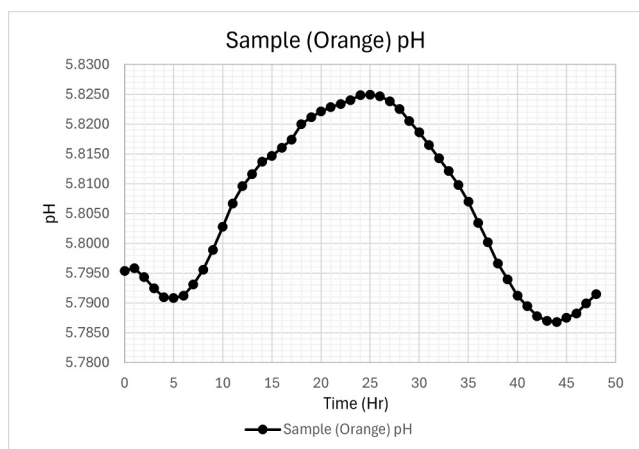


FIGURE 24 | Changes in half orange pH during spoilage.

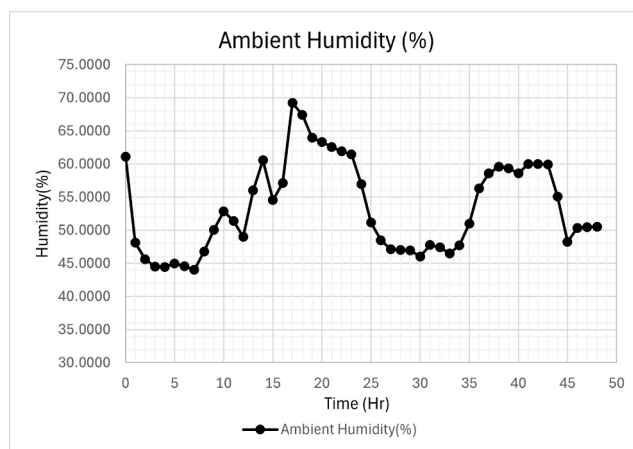


FIGURE 26 | Orange, egg, meat influencing container humidity.

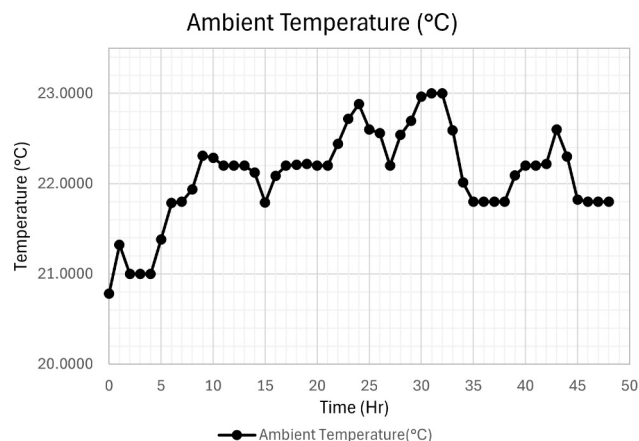


FIGURE 25 | Orange, egg, meat influencing container temperature.

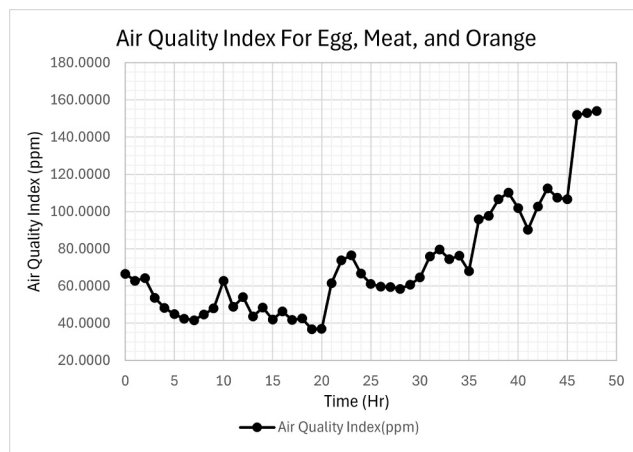


FIGURE 27 | Orange, egg, meat AQI measurement.

4.4 | Machine Learning Model Discussion

If a machine learning model is trained on outputs indicative of spoilage, such as pH or released gases, such as AQI, temperature could accelerate these changes. Incorporating temperature and humidity may help predict when other measurements, such as AQI and pH, become indicative of spoilage. By combining these data, it may be possible to estimate how long a given food item will remain safe to eat at a specific temperature. A predictive model using multi-input regression would be appropriate, as the goal is to monitor data collected by the sensors used in this study and predict the number of hours remaining until the food item becomes unsafe to consume.

4.5 | Potential Ethical and Privacy Concerns

A major consideration in developing such a model is the specificity of the training data. Ethical concerns may arise from conducting several thousand trials that monitor food as it spoils, particularly when animal products, especially those involving muscle tissue are used. The resulting food waste could be quite extreme and morally questionable.

Data could be collected more passively and at scale through the widespread adoption of smart refrigerators in homes and businesses. These refrigerators would monitor the environment inside the cooler-box, and users would help train the model by reporting spoilt food items. Essentially, users would label the data collected by the refrigerator, which would then upload the data along with the associated labels, creating a dataset that does not rely on intentional waste. The downside is that such a system would offer minimal benefit to users until the model is well trained.

The refrigerators would require an internet connection, one concern is connected devices may present a network vulnerability, potentially providing unwanted access into homes and businesses. Another is privacy, the devices would be collecting information about buying or eating habits, and maybe be used to target individuals in unforeseen ways.

4.6 | Android Application

Figure 28 shows the Android application developed for data representation. The first screen shot displays the splash screen, showing the application name. The second screen shot illustrates the login and sign-up screens. Users can log in if they have already registered, or sign up if they are new by providing personal information such as their name, email ID, age, and phone number. Additionally, there is the option to log in using Facebook or Google for user convenience.

The dashboard screen includes the following modules: User Profile, Sensor Details/Configuration, Food Checker, Food History, Device Info, and Analytics. Users can view their personal information in the User Profile module. The Device Info section provides hardware details such as the status and model number of the Raspberry Pi and ESP32. The Sensor Details

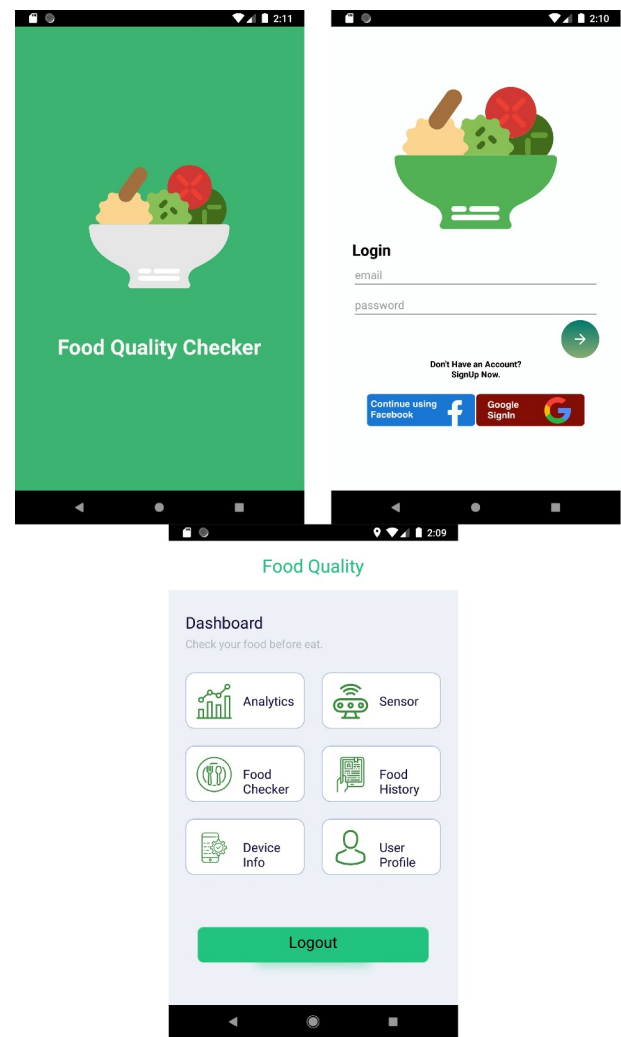


FIGURE 28 | Screenshots of android application.

module shows the status of the sensors, indicating whether they are active or not. The Food History module stores the sensed data from the Raspberry Pi, with individual food IDs for tracking. The Food Checker module displays real-time data for any food, including parameters such as pH, AQI, temperature, humidity, and ammonia contamination. Finally, the Analytics module provides real-time and hourly graphs for each parameter individually.

The Android application was developed using Android Studio on the Windows operating system. Android Studio is an IDE specifically designed for the Google Android operating system, offering an excellent platform for app development and data analysis [56]. The mobile app was written using the Kotlin programming language. Android Studio provides several features, including a fast emulator, an intelligent code editor, and instant app running capabilities.

The proposed model addresses two scenarios. Firstly, consumers are able to detect the state of food items prior to consumption. Secondly, the prototype once scaled and polished, can be utilised in restaurants to monitor all food products, generating a cumulative graph on the store dashboard that assists customers in determining the condition of the food. Existing models are

predominantly applied to industrial food monitoring systems. In contrast, the prototype is designed for personal use by consumers at reduced scale, and individually affordable price. With further development, the prototype can be reduced to a size comparable to a mobile phone, making it extremely portable. By implementing food monitoring measures, the Edispotter can contribute to preventing foodborne illnesses.

5 | Product Analysis

After reviewing existing systems and comparing them with the method discussed in this study, the Edispotter is distinguished by its unique features. The primary advantage of the Edispotter is its consideration of multiple factors that impact food quality. As previously discussed, the freshness of food is influenced by temperature, humidity, pH, and AQI. In daily life, consumers will eat most foods based on assumptions, only discarding it if its obviously spoilt. With the Edispotter, customers can assess the freshness of any food item before consumption in restaurants, contributing to a healthier and more informed lifestyle. Additionally, an Android application provides real-time data for each food item, taking all relevant factors into account. AWS storage is utilised to manage large amounts of data. However, it is important to note that an internet connection is required to enable the Raspberry Pi and its sensors to send data to the Android application for food analysis.

Food hygiene has become increasingly important, making systems designed to check the freshness of food valuable. The Edispotter aims to provide real-time analysis to users, helping to reduce the health risks associated with consuming spoilt food. If a policy were implemented by the government, requiring food industries to display real-time data on factors such as temperature, humidity, pH level, and air quality to customers, consumers would be able to evaluate the freshness of food items before consumption. This would empower consumers to make informed decisions about whether a food item is desirable. Popular fast food restaurants such as Tim Hortons, Starbucks, KFC, McDonald's, and others, could leverage this system to enhance customer confidence by ensuring that the food they purchase is indeed fresh. Additionally, this product could be utilised at the table, allowing customers to conveniently check the freshness of food as it arrives. The system could also be employed by kitchen staff to verify the quality of food before it is served to customers.

The cost of the Edispotter is determined by the hardware and cloud storage purchased from AWS. The total cost of the equipment is approximately 600 CAD, which includes the Raspberry Pi, ESP32 microcontroller, various sensors, and the cloud instance running on AWS.

6 | Conclusion

Foodborne illnesses are a significant concern for public health, and the technology behind food quality monitoring is making a noticeable impact. After reviewing existing systems and exploring various research articles, it became clear that technology can play a crucial role in improving food safety. This

paper presents a prototype called the 'Edispotter', which includes an Android app designed to display food quality readings both graphically and numerically, allowing for easy remote monitoring. The goal of the Edispotter is to reduce the health risks associated with consuming spoilt or unsafe food that may otherwise appear edible. Integrating this system into daily life will help people make healthier food choices and protect against potential health threats and such as foodborne diseases.

Author Contributions

Rashmi Tailor: conceptualization, formal analysis, methodology, resources, software, writing – original draft. **Smit Parikh:** conceptualization, formal analysis, methodology, software, writing – original draft. **Kamalakannan Kumar:** conceptualization, formal analysis, methodology, resources, software, writing – original draft. **Hosam El-Ocla:** formal analysis, supervision, writing – review and editing. **Thomas Collins:** resources, software, validation, visualization, writing – review and editing.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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